16 There is a large number of cubes in a bag.
Jason wants to work out an estimate for the number of cubes in the bag.

He takes at random 10 cubes from the bag.
He puts a mark on each cube and then puts each cube back in the bag.

Jason shakes the bag and then takes at random 20 of the cubes.
There is a mark on 3 of the cubes.

Work out an estimate for the total number of cubes in the bag.

1 mark
Use sample proportion

Mark	All
3	20
10	___

Mark = marked cubes.

All = total cubes.

$$\frac{10}{N} = \frac{3}{20}$$

1 mark
Form proportion

$N = 10 \times 20 \div 3 = 66.666\ldots \approx 67$ cubes.

1 mark
Estimate67

(Total for Question 16 is 3 marks)